

PATENT APPLICATION

Title: System and Method for Domain-Agnostic Hierarchical Autonomous Control Using Dual-Transformer Predictive-Reactive Architecture with Mathematical Verification and Closed-Loop Optimization

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FIELD OF THE INVENTION

This invention relates to autonomous control systems, and more particularly to a hardware-agnostic, four-component hierarchical architecture comprising a predictive sequence transformer, a reactive execution transformer, a mathematical verification engine, and a closed-loop optimization layer. The system is engineered to provide mathematically verified, physically safe control across natural language processing, cybersecurity, robotics, autonomous vehicle control, and financial systems without modification to the core architectural components.

BACKGROUND OF THE INVENTION

Existing autonomous AI architectures suffer from a common structural deficiency: the prediction of an optimal next action and the physical or digital execution of that action are performed by a single monolithic model without an intermediate, isolated mathematical verification step. This structural deficiency produces physical and computational failure modes across deployed systems.

First, unverified generative outputs pose severe hardware and operational risks. A monolithic model may hallucinate a plausible but operationally invalid output, such as a steering command that violates physical kinematic limits, a robotic actuation request that miscalculates the center of gravity, or a firewall rule that induces infinite routing loops. Executing these outputs directly leads to the degradation or destruction of physical server hardware, robotic actuators, and corresponding infrastructure.

Furthermore, existing continuous-monitoring architectures suffer from catastrophic computational bottlenecks. Monitoring thousands of parallel network or sensor streams using standard transformer self-attention mechanisms scales quadratically ($O(N^2)$). This rapidly exhausts available Graphical Processing Unit (GPU) memory allocation, initiating thermal throttling and rendering real-time edge-compute deployments physically impossible without massive, localized server farms.

Second, existing models exhibit non-interpretability of failure. When a monolithic model produces a physically harmful output, the causal chain from input to output is not inspectable. Engineers cannot determine which component of the model's internal state produced the error, making systematic hardware-level correction impossible.

Third, existing systems utilize static optimization. They do not systematically capture rejected outputs as training signals. Each failure is discarded rather than converted into domain-specific training data, preventing the control system from adapting to the specific operational friction of the physical environment in which it is deployed.

The present invention addresses these failure modes through the architectural separation of prediction, execution, verification, and optimization into four distinct, independently operable components connected by mathematically defined, hardware-agnostic interfaces.

SUMMARY OF THE INVENTION

The present invention provides a four-component autonomous control system configured to securely govern physical hardware and digital environments. The system comprises:

A first computational component, the Predictive Transformer **(100)**, which receives a sequence of input state vectors **(101)** representing the historical behavior of a monitored system **(600)** and produces a target state vector **(102)** representing the predicted optimal future thermodynamic or operational state.

A second computational component, the Reactive Transformer **(200)**, which receives the target state vector **(102)** from the Predictive Transformer **(100)** and generates one or more candidate action representations **(201)** designed to drive the monitored system **(600)** toward the target state.

A third computational component, the Verification Engine **(300)**, which acts as a mathematical safety boundary. The Verification Engine **(300)** receives each candidate action **(201)** from the Reactive Transformer **(200)**, applies it to a Digital Twin **(310)** representing the monitored system's current physical or network state, and produces a simulated future state vector. It evaluates whether the simulated future state satisfies physical and operational acceptance criteria defined by the Predictive Transformer **(100)**. Candidate actions **(201)** whose simulated outcomes violate safety envelopes or kinematic constraints are rejected without deployment to the monitored system **(600)**, physically safeguarding the hardware.

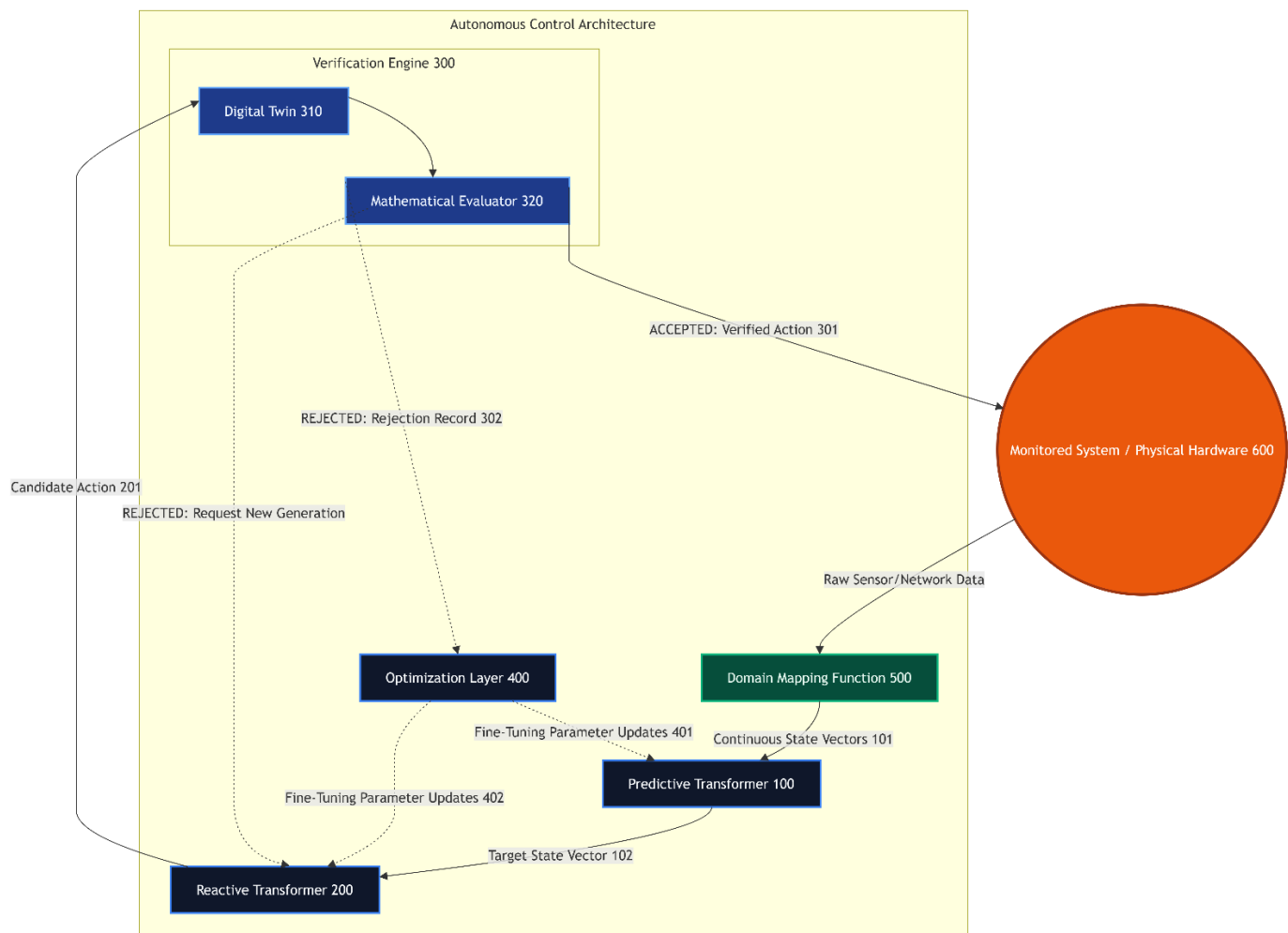
A fourth computational component, the Optimization Layer **(400)**, which receives all rejected candidate actions and their associated rejection records **(302)** from the Verification Engine **(300)**. It logs them as structured training records and periodically applies them to update the parameters **(401, 402)** of the Predictive Transformer **(100)** and Reactive Transformer **(200)** through a closed-loop fine-tuning procedure.

The architecture inherently provides a technical solution to GPU memory exhaustion: it operates as a dual-tier macro/micro hybrid system, utilizing the Predictive Transformer **(100)** to monitor global environments and instantiating highly localized, linear ($O(N)$) sequence models exclusively when an anomaly is verified, thereby drastically reducing the computational memory footprint required for edge-compute hardware.

The architecture is entirely domain-agnostic. The Predictive Transformer **(100)** and Reactive Transformer **(200)** operate exclusively on continuous numerical vectors. Domain specificity is introduced exclusively through a Domain Mapping Function **(500)** that converts domain-specific raw inputs into the standardized vector representation consumed by the system.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating the dual-transformer predictive-reactive architecture, depicting the flow of continuous state vectors through the four core computational components, the Domain Mapping Function, and the continuous feedback loop initiated by the Verification Engine upon rejection of a candidate action.



DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

4.1 The Predictive Transformer (Brain 1)

The Predictive Transformer **(100)** operates as a sequence model trained to predict the optimal next state of a monitored system **(600)** given a fixed-length window of prior states. Each state is represented as a continuous numerical vector of dimensionality D , strictly bounded to a 26-value coordinate matrix ($\mathbf{S} \in \mathbb{R}^{26}$). This specific 26-value constraint is a critical architectural feature that preserves spatial mapping consistency while mathematically preventing geometric overflow or the processing of invalid inputs.

In the preferred embodiment, the Predictive Transformer **(100)** functions as a thermodynamic physics engine, calculating the operational "friction" of a given trajectory. It receives a tensor of shape $[window_size, 26]$ representing historical inputs **(101)** and produces a target state vector **(102)** representing the predicted optimal future state (i.e., the state of lowest thermodynamic friction or highest operational equilibrium).

The Predictive Transformer **(100)** operates in two primary states: an *Observer State* (monitoring mode) where it continuously calculates vector trajectories at low computational overhead, and an *Admin State* (alarm mode), triggered when internal friction calculations indicate an impending breach of operational boundaries, thereby forcing a transition to the Verification Engine **(300)**.

4.2 The Reactive Transformer (Brain 2)

The Reactive Transformer **(200)** is an independently trained execution model (typically instantiated as a localized Small Language Model or Control Transformer) that receives the target state vector **(102)** from the Predictive Transformer **(100)**. Its sole function is to generate a configurable number of candidate action representations **(201)** designed to drive the monitored system **(600)** toward the target state.

Crucially, the Reactive Transformer **(200)** does not possess direct execution authority over the monitored system **(600)**. In robotics or autonomous vehicle applications, it generates numerical control commands (e.g., steering torque, actuator voltage); in software environments, it generates

syntactic code or dialogue. All outputs are strictly quarantined as candidate actions **(201)** pending mathematical validation.

4.3 The Verification Engine (Brain 3) and Mathematical Constraints

The Verification Engine **(300)** is the core safety mechanism of the architecture, physically isolating the Reactive Transformer **(200)** from the monitored hardware **(600)**. It maintains a Digital Twin **(310)** representing the exact current state of the monitored system.

To satisfy the requirement of mathematically preventing invalid inputs, the Mathematical Evaluator **(320)** applies a strict bounding logic preserving spaces up to exactly 26 values. Let the simulated post-action state vector be defined as $\mathbf{S}_{sim} = [s_1, s_2, \dots, s_{26}]$. For any candidate action **(201)**, the Verification Engine **(300)** calculates the simulated Kinetic Envelope (K_e).

The operational friction (F_{total}) introduced by the candidate action is calculated as the weighted sum of divergences from the Predictive Transformer's target state ($\mathbf{S}_{target}$):

$$F_{total} = \sum_{i=1}^{26} w_i |s_{target,i} - s_{sim,i}|$$

Where w_i represents the domain-specific criticality weight for the i -th coordinate. The Mathematical Evaluator **(320)** enforces hard operational constraints defined at deployment time. If any spatial value s_i in the simulated state exceeds the defined safe operational boundary ($B_{max,i}$) or falls below ($B_{min,i}$), the input is instantly mathematically invalidated:

$$\text{If } \exists i \in \{1, 2, \dots, 26\} \text{ such that } s_{sim,i} \notin [B_{min,i}, B_{max,i}]$$

If this condition is met, the candidate action **(201)** is immediately classified as REJECTED. This mathematical cage ensures that an autonomous vehicle cannot execute a steering angle that violates the physical friction coefficient of its tires, and a robotic chassis cannot execute an actuator command that forces its center of gravity outside its base of support. Only candidate actions that satisfy all 26 spatial value constraints and minimize F_{total} are marked as ACCEPTED **(301)** and deployed to the hardware **(600)**.

4.4 The Optimization Layer (Brain 4)

The Optimization Layer **(400)** provides a bidirectional closed-loop system for artificial homeostasis and continual learning. It receives structured rejection records **(302)** from the Verification Engine **(300)** when a candidate action violates the 26-value boundary constraints.

First, it maintains an auditable, persistent log of these rejections, allowing human administrators to inspect causal failure chains without deciphering opaque neural network weights. Second, it periodically leverages these logged rejection records **(302)** as negative-constraint training data. Utilizing parameter-efficient techniques such as Low-Rank Adaptation (LoRA) or Quantized LoRA (QLoRA), it mathematically updates the weights **(401, 402)** of the Predictive Transformer **(100)** and Reactive Transformer **(200)** entirely on local hardware. This ensures the system autonomously learns the physical or operational boundaries of its specific deployment environment without requiring external cloud computation.

4.5 The Domain Mapping Function

The Domain Mapping Function **(500)** is the sole modular component that changes when the architecture is deployed across disparate industries. Its purpose is to ingest domain-specific raw inputs (e.g., optical camera feeds, LIDAR point clouds, natural language text, or network traffic packets) and translate them into the standardized 26-value continuous numerical vectors **(101)** required by the Predictive Transformer **(100)**.

In an autonomous vehicle embodiment, the mapping function extracts kinematic variables (yaw, pitch, roll, acceleration, proximity) into the 26-value array. In a conversational or corporate AI embodiment, the mapping function utilizes an embedded classifier to map syntactic intent, semantic relationships, and relational trust scores into the exact same 26-value geometric array. Consequently, the core four-brain architecture processes vehicle kinematics and human dialogue using identical thermodynamic mathematics.

4.6 Macro/Micro Hybrid Memory Allocation

To overcome the quadratic computational overhead ($O(N^2)$) inherent to standard transformer self-attention in continuous-monitoring environments, the architecture implements a dual-tier, hardware-optimized memory allocation method. The Predictive Transformer **(100)** operates

initially as a global macro-sensor, aggregating total system traffic to detect overarching trajectory shifts with minimal compute.

Upon calculating a target state vector **(102)** that breaches a defined friction threshold, the system dynamically instantiates transient, localized micro-sequence models operating at linear computational complexity ($O(N)$). These micro-models isolate specific entity interactions (e.g., a single physical actuator or a single network IP address sequence) to verify the anomaly, and are purged from memory immediately upon equilibrium restoration. This macro-to-micro handoff drastically reduces GPU memory allocation, guaranteeing that the control architecture can run locally on resource-constrained edge-compute nodes integrated directly into physical hardware.

CLAIMS

Claim 1. A computer-implemented system for the autonomous control of a localized hardware or network environment, the system comprising:

at least one physical processor; and

a non-transitory computer-readable memory storing instructions that, when executed by the at least one physical processor, instantiate a predictive-reactive architecture comprising:

a first computational component, the Predictive Transformer, configured to receive a sequence of input state vectors representing historical observations of a monitored physical or network system, and to produce a target state vector representing a predicted optimal future state of the monitored system;

a second computational component, the Reactive Transformer, configured to receive the target state vector from the first component and to generate a plurality of candidate action representations, each candidate action representing a proposed physical or digital intervention on the monitored system;

a third computational component, the Verification Engine, configured to apply each candidate action to a mathematical model of the monitored system's current state, produce a simulated future state vector, submit the simulated future state vector to the first component for trajectory

evaluation, and classify each candidate action as ACCEPTED if and only if the predicted trajectory from the simulated state satisfies a strict set of geometric and kinematic acceptance criteria, or REJECTED otherwise, wherein no REJECTED candidate action is deployed to the monitored system; and

a fourth computational component, the Optimization Layer, configured to receive structured rejection records from the third component comprising each rejected candidate action, to maintain a persistent human-readable log of said rejection records, and to periodically apply accumulated records to update the parameters of the first and second components through a localized fine-tuning procedure;

wherein the third component mathematically verifies all candidate actions prior to deployment to physically safeguard the monitored system and prevent the execution of resource-destructive commands on connected physical hardware.

Claim 2. The system of Claim 1, wherein the continuous numerical vectors utilized by the first, second, and third components are strictly bounded to a 26-value geometric array, and wherein the third component mathematically prevents the processing of invalid inputs by rejecting any candidate action whose simulated outcome violates defined operational boundaries within the 26-value spatial constraint.

Claim 3. The system of Claim 1, wherein the acceptance criteria evaluated by the third component comprise calculating a thermodynamic operational friction introduced by the candidate action, ensuring the target state specified by the first component is reached within a configurable number of future time steps, and ensuring no intermediate state in the predicted trajectory violates hard operational constraints defined at deployment time.

Claim 4. The system of Claim 1, wherein the third component is further configured, upon finding no ACCEPTED candidate action among the plurality generated by the second component, to return all rejection records to the second component as negative context and request a subsequent generation cycle, up to a configurable maximum number of cycles.

Claim 5. The system of Claim 1, further comprising a Domain Mapping Function configured to convert domain-specific raw inputs into continuous numerical vectors, wherein the Domain Mapping Function is the sole modular component that changes between deployment domains, thereby ensuring the first, second, third, and fourth components operate exclusively on continuous vectors and remain architecturally unchanged across disparate industries.

Claim 6. The system of Claim 5, wherein the Domain Mapping Function, when applied to network traffic inputs, applies a deterministic algorithmic parser to extract features comprising: the ratio of connection-initiation packets to completed handshakes per unit time; the ratio of outbound to inbound data volume per session; the frequency of protocol anomalies relative to an established baseline rate; and the proportion of traffic falling outside all defined protocol categories.

Claim 7. The system of Claim 5, wherein the Domain Mapping Function, when applied to autonomous vehicle sensor inputs, extracts kinematic features comprising position, velocity, acceleration, and heading of a controlled vehicle and all detected objects within a defined radius, mapping them directly into the 26-value continuous vector array.

Claim 8. The system of Claim 1, wherein the Optimization Layer updates the parameters of the first and second components utilizing parameter-efficient Low-Rank Adaptation (LoRA) or Quantized Low-Rank Adaptation (QLoRA) performed exclusively on local enterprise hardware, removing dependency on external cloud computation.

Claim 9. The system of Claim 1, further comprising a macro/micro hybrid memory allocation architecture, wherein the first component evaluates aggregate system data using a global attention mechanism operating at quadratic computational complexity ($O(N^2)$), and, upon predicting a target state vector that breaches an anomaly threshold, dynamically instantiates one or more transient sequence models operating at linear computational complexity ($O(N)$) to isolate and track specific entity interactions, thereby preventing Graphical Processing Unit (GPU) memory exhaustion on edge-compute hardware.

Claim 10. A computer-implemented method for the physically safe autonomous control of a monitored physical or digital system, the method comprising the steps of:

receiving, at a first computational component, a sequence of input state vectors representing historical observations of the monitored system;

producing a target state vector representing a predicted optimal future state;

generating, at a second isolated computational component, a plurality of candidate action representations designed to drive the monitored system toward the target state;

intercepting the candidate actions at a mathematical Verification Engine prior to deployment;

applying each candidate action to a Digital Twin of the monitored system's current state to produce a simulated future state vector;

evaluating whether the simulated future state violates strict 26-value geometric and kinematic constraints;

deploying to the monitored system only candidate actions whose simulated outcomes satisfy all mathematical acceptance criteria;

logging all rejected candidate actions with their associated mathematical rejection reasons; and

periodically applying logged rejection records to locally update parameters of the first and second computational components.

Claim 11. The method of Claim 10, wherein the steps are performed without architectural modification to the core computational components when applied to operational domains comprising natural language processing, network security monitoring, autonomous vehicle control, robotic systems control, and financial market systems.

ABSTRACT

A hardware-agnostic, four-component autonomous control system is disclosed. A Predictive Transformer **(100)** receives continuous state vectors **(101)** representing a monitored physical or network system **(600)** and predicts an optimal target state vector **(102)**. An isolated Reactive Transformer **(200)** generates candidate actions **(201)** to achieve the target state. A Verification Engine **(300)** intercepts the candidate actions, simulates them against a Digital Twin **(310)**, and mathematically rejects actions violating a strict 26-value spatial constraint boundary, physically safeguarding the system from destructive generative hallucinations. An Optimization Layer **(400)** converts rejected actions **(302)** into localized training data to continuously fine-tune the transformers **(100, 200)** via closed-loop feedback. Domain specificity is introduced via a Domain Mapping Function **(500)**, preserving the core architecture across natural language, robotics, and edge-compute applications without modifying the underlying vector mathematics.